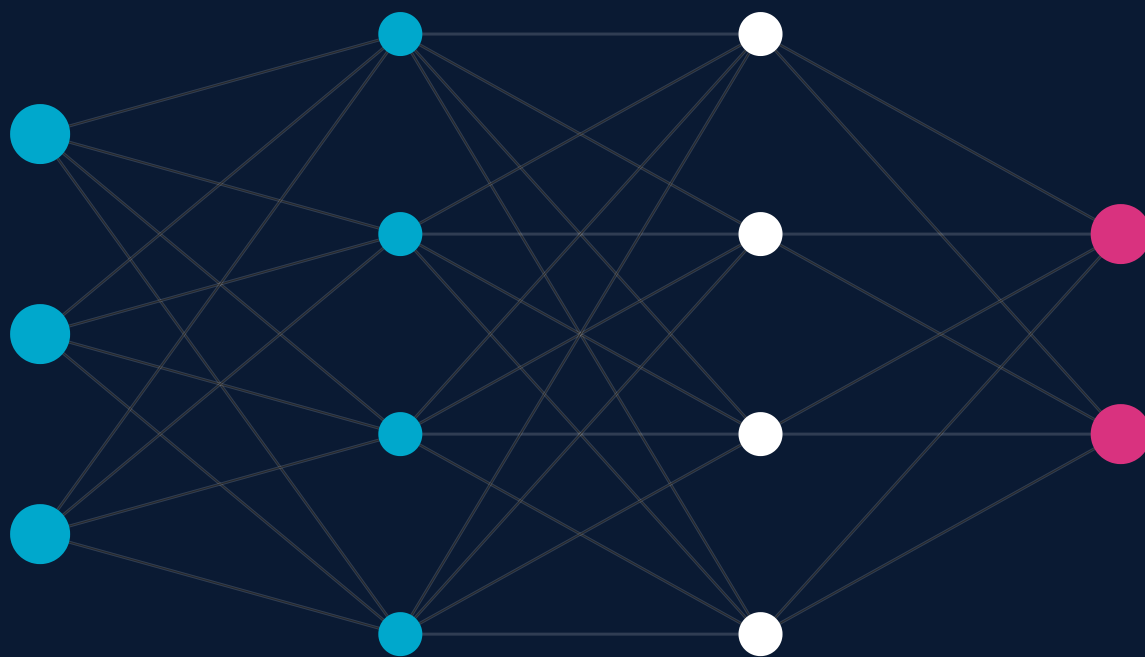




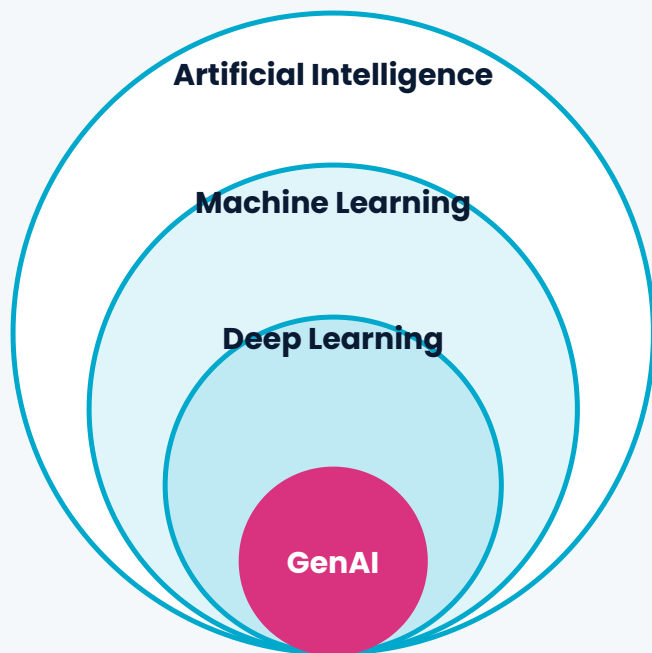
AI & ML Career Starter

Skills. Roadmap. Projects.

Understand the skills, roadmap, projects, and foundations needed to begin an AI and ML career.

What are AI & ML?

Know the landscape before you pick a path.



● Artificial Intelligence

Machines doing tasks that normally need human intelligence.

● Machine Learning

Systems that learn patterns from data instead of fixed rules.

● Deep Learning

Machine learning with many-layered neural networks.

● Generative AI

Models that create text, images, code and more.

HOW AI WORK FLOWS

01

Define the problem

02

Collect and clean data

03

Train models

04

Evaluate results

05

Deploy and monitor

IS THIS CAREER FOR YOU?

- ☐ You enjoy solving problems step by step.
- ☐ You are curious about data and patterns.
- ☐ You are happy to keep learning new tools.
- ☐ You can practise regularly, even 1 hour a day.

GOOD NEWS

You do not need a PhD to start in many entry-level data and AI roles. You do need solid foundations and real projects.

Roles in AI & ML

Pick a first target. You can move between roles later.

Data Analyst

Find insights in data and share them clearly.

SQL

Excel

Visualisation

Statistics

HOW HARD TO START



Data Scientist

Use data and models to answer business questions.

Python

Statistics

ML

Communication

HOW HARD TO START



AI Engineer

Build products using existing models and APIs.

Python

APIs

Prompting

RAG

HOW HARD TO START



ML Engineer

Build and ship machine learning systems.

Python

ML

Software engineering

Deployment

HOW HARD TO START



MLOps Engineer

Keep models running reliably in production.

Cloud

Docker

CI/CD

Monitoring

HOW HARD TO START



Research Scientist

Invent new methods and publish findings.

Advanced maths

Papers

Experiments

HOW HARD TO START



A SMART START

Many people begin as a data analyst or software engineer, then grow toward machine learning.

KEEP IN MIND

Job titles and requirements vary between companies. Read several job posts to see the pattern.

Programming & tools

The everyday toolkit of every AI and ML professional.

SKILLS TO LEARN (DOTS = PRIORITY AT THE START)

Python fundamentals ●●●●●

Variables, loops, functions, classes, files and errors.

NumPy and Pandas ●●●●●

Arrays, DataFrames, filtering, grouping and merging.

Data structures ●●●●○

Lists, dictionaries, sets and when to use each.

Git and GitHub ●●●●○

Commit, branch, push and write a README.

Visualisation ●●●○○

Charts with Matplotlib and Seaborn.

Terminal and environments ●●●○○

Command line, pip, virtual environments, Jupyter.

explore.py

```
1 import pandas as pd
2
3 df = pd.read_csv("data.csv")
4 print(df.head(3))
5 print(df["price"].mean())
6
```

OUTPUT

	size	price
0	50	120
1	80	180
2	65	150
		170.0

TOOLS

Python 3

Jupyter

VS Code

Git

Google Colab

MINI-TASK

Load any CSV with Pandas, print the first rows and plot one chart.

Maths & statistics

The ideas behind how models learn. Intuition first.

SKILLS TO LEARN (DOTS = PRIORITY AT THE START)

Linear algebra



Vectors, matrices and dot products.

Why it matters: data and model weights are stored as matrices.

Statistics



Mean, variance, sampling, hypothesis tests, correlation.

Why it matters: you must judge whether results are real.

Probability



Distributions, conditional probability, Bayes' rule.

Why it matters: models always deal with uncertainty.

Calculus



Derivatives, gradients and the chain rule.

Why it matters: training adjusts weights using gradients.

math_basics.py

```
1 import numpy as np
2
3 a = np.array([1, 2, 3])
4 b = np.array([4, 5, 6])
5 print(np.dot(a, b))
6 print(a.mean(), a.var())
7
```

OUTPUT

```
32
2.0 0.6666666666666666
```

HOW DEEP DO YOU NEED TO GO?

- ☐ **Start:** understand ideas and intuition.
- ☐ **Then:** apply them in code.
- ☐ **Later:** go deeper if your path needs it.

MINI-TASK

Learn maths alongside projects, not before them. Compute a mean, variance and dot product yourself.

Data skills

Models are only as good as the data behind them.

SKILLS TO LEARN (DOTS = PRIORITY AT THE START)

SQL



SELECT, WHERE, JOIN, GROUP BY.

Data cleaning



Missing values, duplicates, wrong types, outliers.

Exploratory analysis



Summaries, distributions and correlations.

Data storytelling



Explain findings clearly to non-experts.

Feature engineering



Encoding, scaling and creating new features.

query.sql

```
1 SELECT department, AVG(salary)
2 FROM employees
3 GROUP BY department;
4
```

OUTPUT

```
('Data', 100.0)
('Sales', 75.0)
```

A LARGE SHARE OF ML WORK IS DATA WORK

Clean, well-understood data usually improves results more than a fancier model.

MINI-TASK

Pick a public dataset, clean it and write 3 findings in plain language.

Machine learning

Your first model, the workflow and the ideas that matter.

01

Frame the problem

02

Prepare data

03

Choose a model

04

Train

05

Evaluate

06

Improve and deploy

MAIN TYPES OF TASK

Regression Linear regression, random forest Predict house prices

Classification Logistic regression, decision trees Spot spam emails

Clustering k-means Group similar customers

Dimensionality reduction PCA Simplify many features

Iris is a tiny, easy dataset. Real data is messier, so expect lower scores.

```

first_model.py

1 from sklearn.datasets import load_iris
2 from sklearn.model_selection import train_test_split
3 from sklearn.ensemble import RandomForestClassifier
4
5 X, y = load_iris(return_X_y=True)
6 X_tr, X_te, y_tr, y_te = train_test_split(
7     X, y, test_size=0.2, random_state=42)
8 model = RandomForestClassifier(random_state=42)
9 model.fit(X_tr, y_tr)
10 print("Accuracy:", model.score(X_te, y_te))
11

```

OUTPUT

Accuracy: 1.0

Train / test split

Keep unseen data to test honestly.

Overfitting

Great on training data, poor on new data.

Metrics

Accuracy, precision, recall, F1, RMSE. Pick by problem.

MINI-TASK

Run this code, then change `test_size` and a model setting. Note what happens to the score.

Deep learning & LLMs

Learn the basics, then specialise where you are curious.

SKILLS TO LEARN (DOTS = PRIORITY AT THE START)

Neural network basics ●●●●●

Layers, weights, activation, loss and gradient descent.

Common architectures ●●●●○

CNNs for images, RNNs for sequences, Transformers for language and more.

LLM concepts ●●●●○

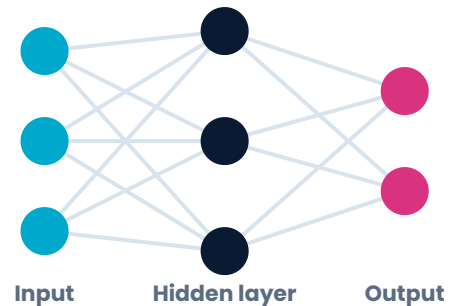
Tokens, prompts, context window, embeddings.

Applying models ●●●○○

Fine-tuning, RAG (retrieve documents, then generate) and evaluation.

Frameworks ●●●●○

PyTorch or TensorFlow/Keras, plus Hugging Face.



TOOLS

PyTorch

Keras

Hugging Face

Google Colab

Kaggle Notebooks

USE AI RESPONSIBLY

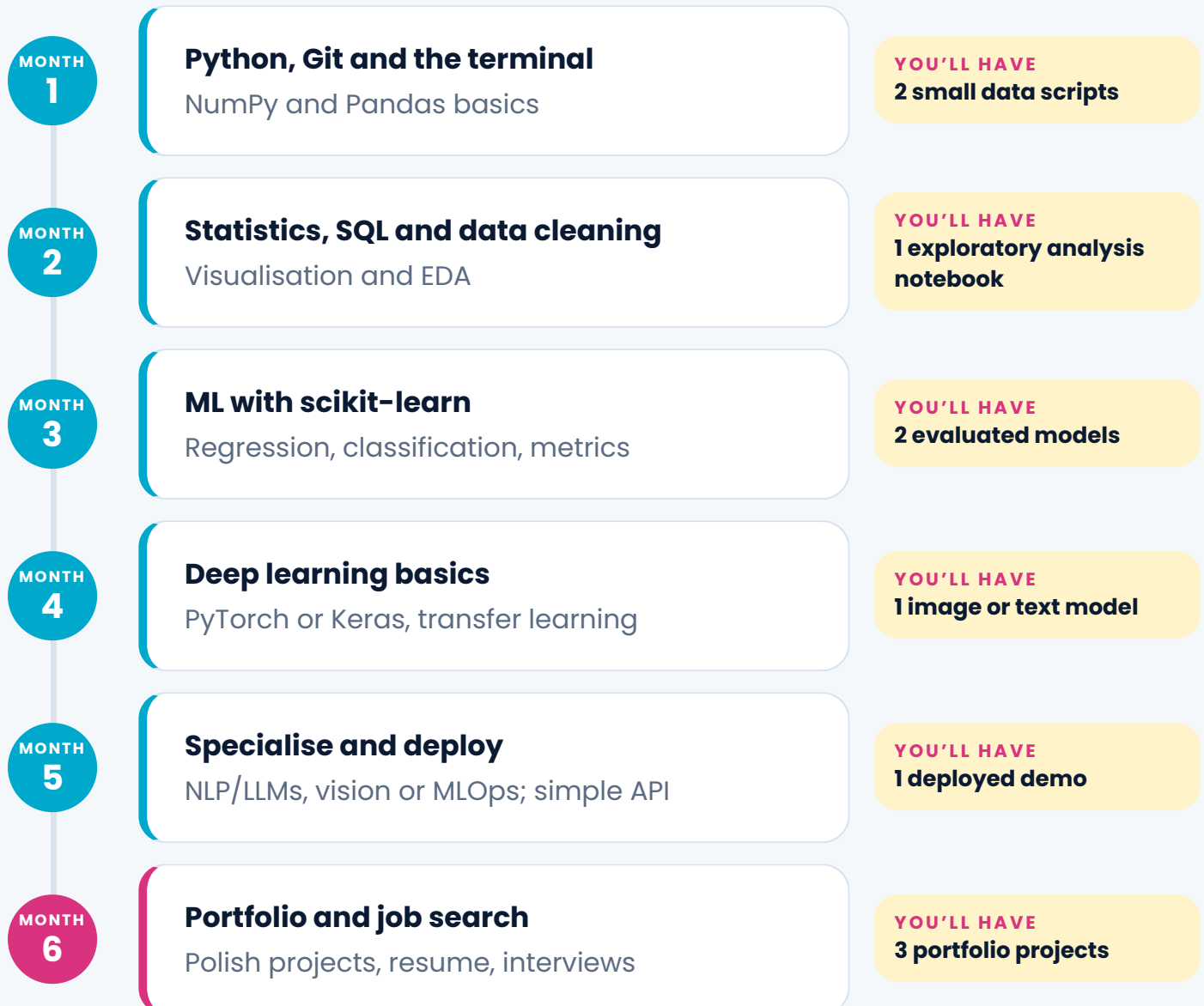
Check data for bias, protect privacy, test for wrong or invented answers, and be clear about what a model can and cannot do.

MINI-TASK

Use a pretrained image model to classify 10 of your own photos. Note where it is wrong.

Your 6-month roadmap

A realistic plan at about 10 to 15 hours a week. Adjust to your pace.



Build a project ladder

Start small. Each level adds one new skill.

Data detective

LEVEL
1

Explore and clean a public dataset. Share 3 clear findings with charts.

DIFFICULTY



Pandas

EDA

Storytelling

Price predictor

LEVEL
2

Predict house or car prices with regression. Compare 2 models.

DIFFICULTY



Regression

Metrics

scikit-learn

Classifier

LEVEL
3

Spot spam, predict customer churn or classify sentiment.

DIFFICULTY



Classification

Precision / recall

Feature work

Image classifier

LEVEL
4

Fine-tune a pretrained model on your own image set.

DIFFICULTY



Deep learning

Transfer learning

PyTorch / Keras

End-to-end app

LEVEL
5

Deploy a model API or a document Q&A chatbot with a simple interface.

DIFFICULTY



Deployment

LLMs / RAG

Software skills

EVERY PROJECT NEEDS

- ☐ A clear problem statement
- ☐ A README: problem, data, method, results
- ☐ Code others can run
- ☐ Public data, with the source named
- ☐ Honest results and limitations
- ☐ A demo or screenshots

Portfolio & interviews

Show your work. Explain your thinking.

PORTFOLIO CHECKLIST

- ☐ GitHub profile with 3 to 5 pinned projects
- ☐ Each repo has a clear README
- ☐ A short write-up or notebook per project
- ☐ A resume that links to your best work
- ☐ An up-to-date LinkedIn profile
- ☐ Optional: a Kaggle profile or short blog

RESUME BULLET FORMULA

Built [what] **using** [tools] **to** [result].

Example: Built a churn classifier with a random forest and tested it on held-out data, reaching 0.84 F1.

PREPARE FOR 4 AREAS

Python and SQL

"Find the top 3 customers by revenue."

ML fundamentals

"What is overfitting? How do you reduce it?"

Metrics

"Explain precision vs recall."

Your projects

"Walk me through your best project and why you chose that model."

WHERE TO MEET PEOPLE

GitHub

Kaggle

LinkedIn

Local meetups

Open-source projects

Online communities

REMEMBER

Interviewers care more about how you think and explain than about memorised definitions.

Where are **you now?**

Fill the dots (1 to 5), then focus on your lowest scores.

SKILL	MY LEVEL	NEXT ACTION
Python	○ ○ ○ ○ ○	_____
NumPy and Pandas	○ ○ ○ ○ ○	_____
SQL	○ ○ ○ ○ ○	_____
Statistics and maths	○ ○ ○ ○ ○	_____
Machine learning	○ ○ ○ ○ ○	_____
Deep learning	○ ○ ○ ○ ○	_____
Projects and GitHub	○ ○ ○ ○ ○	_____
Communication	○ ○ ○ ○ ○	_____

YOUR FIRST 30 DAYS

Week 1

Install Python and Git. Finish a Python basics course.

Week 2

Learn Pandas. Analyse one dataset.

Week 3

Learn SQL basics. Write 10 queries.

Week 4

Train your first scikit-learn model. Post it on GitHub.

RESOURCES TO EXPLORE

[Andrew Ng's Machine Learning Specialization](#)
[fast.ai Practical Deep Learning](#)
[Kaggle Learn courses](#)
[Google Machine Learning Crash Course](#)
[Hugging Face courses](#)
[3Blue1Brown linear algebra videos](#)

Course availability and pricing change, so check each site. This guide offers general career guidance, not a promise of any job outcome.